Round 175 Diad Discovery + Discipline Withdrawal (Backbone-First, F_TRZ-Audit-Guided): (1) B/B_crit(Sombrero Superconductivity) = 1e-10/1e11 = 1e-21 = F_TRZ^21 EXACT NEW Family-Extension Dimensional-Object Application of F_TRZ^21 Rung to Magnetic-Field-Ratio Domain at Sombrero (F_TRZ^21 Rung Documented in Strain/LIGO PAPER_1989 + CenA/Crab PAPER_1983 But Not at Sombrero Field-Ratio); (2) f_aether(SgrA) = 1e3 Hz = SO_5^3 Hz EXACT NEW 2ND Aether-Frequency Slot (Extends PAPER_2023 R157 D5 f_aether = SO_5^4 Hz Seminal First-Documented Aether-Frequency Slot to 2-Object Family General + SgrA); Plus Discipline Withdrawal omega(SgrA) = F_TRZ^6 rad/s = CROSS-OBJECT PAPER_1994 R132 SMBH Rotation Angular Velocity Seminal (Not R175 Novel — Restated)

Daniel T. Murphy

PAPER_2044 — Round 175 Diad Discovery + Discipline Withdrawal (Backbone-First, F_TRZ-Audit-Guided)

Author: Daniel T. Murphy | **Framework:** UQFF Star-Magic v5.68+ | **Date:** 2026-07-15

Motivation — 34th Consecutive Backbone-First Round (F_TRZ-Audit-Guided)

R175 was guided by PAPER_2043 F_TRZ^n ladder cross-domain population audit. Target: fill audit-identified F_TRZ ladder gaps at field-ratio rungs and angular-frequency subladder rungs. Sharper backbone-first double-check confirms 2 novel + 1 discipline withdrawal.

Abstract

Discovery 1 — B/B_crit(Sombrero Superconductivity) = F_TRZ^21 EXACT — Family-Extension Novel:
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B(Sombrero magnetic field) / B_crit(Schwinger critical) = 1e-10 T / 1e11 T
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= 1e-21
= F_TRZ^21 EXACT
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Novel family-extension. F_TRZ^{21} rung is documented at multiple objects across the corpus:
- **PAPER_1989 R123 D1**: LIGO strain sensitivity h = F_TRZ^{21} = 1e-21 (strain-domain)
- **PAPER_1983 R116**: CenA and Crab objects — F_TRZ^{12} + F_TRZ^{21} same-object anchor
- **PAPER_1955**: SO_5^{21} galactic ladder (different primitive-form)

But **Sombrero magnetic-field-ratio dimensional-domain application at n=21** is R175 D1 NEW. Extends F_TRZ^n field-ratio family:

n	Object	Attribution
16	Quantum measurement collapse	PAPER_1869 seminal
16	M16 superconductor B/B_crit	PAPER_2041 R173 D3
19	Crab superconductor B/B_crit	PAPER_2042 R174 D1
21	**Sombrero superconductor B/B_crit**	**PAPER_2044 R175 D1 NEW**

Field-ratio F_TRZ^n family now **4-rung, 4-object** (M16 n=16 + Crab n=19 + Sombrero n=21 + quantum-measurement n=16) — validates F_TRZ ladder cross-domain population expanding through R173-R175 arc.

Discovery 2 — f_aether(SgrA) = 1e3 Hz = SO_5^3 Hz EXACT — 2ND Aether-Frequency Slot:
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f_aether(Sgr A* aether-mediated resonance frequency) = 1e3 Hz = SO_5^3 Hz
EXACT
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Novel 2nd aether-frequency slot in SO_5-power ladder aether-frequency dimensional domain. Extends PAPER_2023 R157 D5 seminal:

n	Value (Hz)	Object	Attribution
+4	1e4	General aether resonance	PAPER_2023 R157 D5 seminal (first documented)
+3	**1e3**	*Sgr A* aether-mediated resonance	PAPER_2044 R175 D2 NEW**

Aether-frequency family now 2-object across 2 rungs (n=3 SgrA + n=4 general).

Cross-Object Withdrawal (Discipline Catch)

omega(SgrA superconductor) = 1e-6 rad/s = F_TRZ^6 EXACT — **WITHDRAWN from novelty**.
Sharper backbone-first check surfaces PAPER_1994 R132 seminal:
- **PAPER_1994** — "F_TRZ^6 SMBH Rotation Angular Velocity Cross-Domain Instance (New for n=6 Rung)"

The R175 F4 SgrA superconductor omega = F_TRZ^6 is a restatement of PAPER_1994 R132's SgrA SMBH rotation angular velocity F_TRZ^6 (same physical object, same value). Not R175 novel. Discipline catch #5 across the arc.

This also refines R174 (PAPER_2042) D3's claim: Resonance Ug4i omega = F_TRZ^3 was novel for angular-freq DOMAIN opening. R175 D2/D3 attempt to extend to n=6 turns out to be same-object restatement of PAPER_1994. The angular-frequency subladder is confirmed as 2-rung at n=2 + n=3 (Tapestry + Resonance, per PAPER_2042), NOT 3-rung at n=6.

Cross-Object Confirmations (R175 fill wire-ins)

- **B(Sombrero) = F_TRZ$^{1■}$ T** → Amplitude/F_TRZ ladder documented (F1)
- **B_crit = SO_5^{11} T** → PAPER_2013 R150 D8 Schwinger (F1)
- **M(Sombrero) = SO_5^{11} M_sun** → galactic mass ladder (F1)
- **A_osc/omega(Sombrero) = F_TRZ$^{1■}$** → PAPER_1919 F_TRZ ladder + PAPER_1880 MICROSCOPE tier (F3)
- **k(Sombrero) = F_TRZ^{21}** → PAPER_1919 F_TRZ ladder (F3)
- **B/B_crit(SGR1745) = 2·F_TRZ** → PAPER_1944 seminal (F2 already documented)
- **I(SgrA) = SO_5$^{2■}$** → PAPER_1994 R132 seminal (F4 + F5)
- **B_proxy(SgrA) = SO_5■■ T** → PAPER_2022 R156 D3 Crab (F5 cross-object)

All 2 discoveries + 1 withdrawal + 8 confirmations validated at fidelity gate 1444/0.

R142-R175 Trajectory (34 consecutive backbone-first rounds)

Round	Novel	Cross-obj	Notes
R142-R174	163 total	22+9+5	33 rounds + PAPER_2043 F_TRZ audit
R175	**2**	**1 withdrawal + 8**	**34th; F_TRZ audit-guided round with 1 discipline catch**

Wiring Plan (2 dispatches, lowercase keys)

- `b_over_b_crit_sombrero_f_trz_21_field_ratio_4th_rung` → 1e-21
- `f_aether_sgra_so_5_3_2nd_aether_freq_slot` → 1e3

Cross-References

- **PAPER_1919** — F_TRZ Power Ladder (D1 + D2 basis)
- **PAPER_1983** — R116 CenA F_TRZ^12 + F_TRZ^21 anchor (D1 attribution)
- **PAPER_1989** — R123 LIGO strain F_TRZ^21 (D1 attribution)
- **PAPER_1994** — R132 F_TRZ^6 SMBH rotation angular velocity seminal (Discipline withdrawal source)
- **PAPER_2023** — R157 D5 f_aether SO_5^4 Hz seminal (D2 basis)
- **PAPER_2041** — R173 F_TRZ^16 M16 field-ratio (D1 field-ratio family predecessor)
- **PAPER_2042** — R174 F_TRZ^19 Crab field-ratio (D1 predecessor)
- **PAPER_2043** — F_TRZ ladder cross-domain population audit (R175 guidance source)

Conclusion

R175 34th-consecutive backbone-first round yields **2 novel structural findings + 1 discipline withdrawal + 8 cross-object attributions**. Audit-guided round successfully filled audit-identified gap in F_TRZ ladder field-ratio dimensional-domain at n=21, and extended SO_5-power aether-frequency family at n=3.

Cumulative through PAPER_2044: 165 first-pass novel + 22+9+5 confirmations + 4 audit sweeps + 4 self-corrections + 1 backbone-recovery + 5 attribution withdrawals + 7 family-extension attributions + 5 discipline-observation formalizations + 1 30-round arc milestone + 1 7th-prefix-class formalization + 1 F_TRZ-ladder broad-framework validation.

Signature milestones this paper:
- **F_TRZ audit-guided round validates PAPER_2043 methodology** — successfully filled predicted gap
- **F_TRZ field-ratio family now 4-object, 4-rung** (n=16 M16 + n=19 Crab + n=21 Sombrero + n=16 quantum-collapse) — most-populated F_TRZ cross-domain family

- **Aether-frequency family 2-object** (SgrA n=3 + general n=4)
- **Discipline catch #5** across the 34-round arc — validates backbone-first sharper-check methodology

End of PAPER_2044 Draft 1.